

Rugby Intelligent Camera

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Abstract—Camera-based tracking systems are increasingly used in sports to enhance performance analysis and broadcast production. However, existing solutions are typically expensive and complex, making them inaccessible for amateur clubs with limited resources. This project aims to develop a cost-effective and functional camera tracking solution to, in real time, broadcast and capture the statistics of an amateur rugby match. Our proposition relies on a multi-camera setup which is controlled by a trained computer vision model, YOLO (You Only Look Once). The conducted tests show an accuracy of 80

Index Terms—computer vision, rugby, cost-effective, camera tracking, YOLO

I. INTRODUCTION

This document is a model and instructions for L^AT_EX. Please observe the conference page limits.

II. TECHNICAL CHOISES (EXPLAINING HOW WE APPROCHED THE PROBLEM)

A. Maintaining the Integrity of the Specifications

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III. OUR WORK

We can change the order of the project aswell

A. YOLO Model

To track the focus of a rugby match, we agreed that the most important thing we should track is the ball, but in rugby the ball is small and always hidden between the players or very close to their chest, causing a big problem for the model's identification of the ball. In order to solve this problem, we added the class player; this implies that the players will also be identified by the model.

In rugby, the teams usually place themselves in lines of attack and defense, so, normally the point of interest will take

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place where the greatest number of players are. With both classes, we can already create a working tracking model.

The world of computer vision developers is growing and one of the biggest communities is the Roboflow Universe, which we can find a great number of models and datasets. The project Rugby Detection Computer Vision Model [8] was the closest to our project's needs, and the pictures used in this dataset are similar to the images we captured in an INSA AS match.

After testing the model from the project [8] in some images from the captured match (Fig. 1 to Fig. 3), positive results were achieved: the model responds well to the images, identifying a big number of player and the ball. Some problems were: confusing the arbitrator with a player, not identifying all the players, and sometimes not seeing the ball.

Unfortunately, the model [8] isn't accessible to download, so it wouldn't be possible to use in a real time application, because for every frame the Roboflow repository would have to be accessed on the internet, increasing the cost and delay of the image processing.

To solve the presented problem, the project's dataset [8] was cloned and a new AI model was trained, hoping to achieve the same results as the previous. Four models were created, two with yolov8n, one with yolov8s and the last with yolov8m; they were all tested in an image not from the dataset [??? Show images of the test???].

These models aren't as good as the previous one, but they are functional for video analyses and can be used for real-time applications.

IMAGES COMPARISON

To validate our models, we tested it in a video of an INSA AS match, but the power of the available GPU wasn't enough to analyze a nineteen second video (with 1141 frames). So, we couldn't validate our models in a hole video but, analyzing in a smaller video, we can see some problems of players and ball identification, showing how the project would look like in a real-time solution.



Fig. 1. Test image 1, used model from Roboflow [8].



Fig. 2. Test image 2, used model from Roboflow [8].

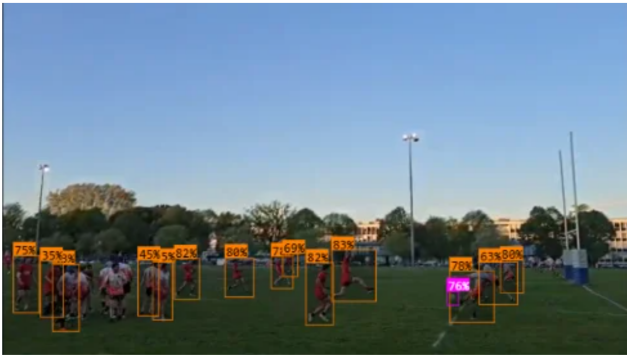


Fig. 3. Test image 3, used model from Roboflow [8].

Regarding the YOLO model we can conclude that this solution works for identification of our needs to choose the best camera but, trying to find a low-cost solution for the amateur clubs we arrived at a lack of computing power that probably will increase the cost of this approach.

To improve this project, we plan on using another YOLO model [9] to see the statistics of the game as well, identifying tackle, penalty kick, throw-in, scrum, and other similar rugby situations that our other model can't identify. Another aimed improvement is to create a system [9] so the model could be able to after identifying the players, separate them between

team A and team B. The system consists of a Sigmoid loss language image pre-trained [9] (extract color information from the player's shirt), associated with an UMAP [9] (compact the previous information) and a Kmeans [9] (sort both teams). These improvements would highly increase the quality of the performance analysis of the teams.

B. Video Positioning teams part

- Use either SI (MKS) or CGS as primary units. (SI units are encouraged.) English units may be used as secondary units (in parentheses). An exception would be the use of English units as identifiers in trade, such as "3.5-inch disk drive".

C. Video Acquisition and Transmission Architecture

Real-time automated tracking of rugby players imposes strict constraints regarding bandwidth, latency, and hardware costs. This section details the technological choices made to ensure a fluid and synchronized video stream to the central processing unit.

D. Capture Device Choice

Several capture solutions were evaluated to meet the requirements of the multi-camera tracking project. The use of cameras on motorized platforms was initially considered for its dynamic tracking capability—which could have reduced the number of cameras needed and thus the total cost. However, extreme sensitivity to control delays and prohibitive costs led to its rejection. Similarly, solutions based on microcontrollers such as Arduino, while inexpensive, offered insufficient acquisition performance and resolution for exploitation via YOLOv8 detection models.

Ultimately, smartphones were selected as the capture units. These devices offer an optimal compromise: label=•

- **Image Quality** : High-resolution sensors capable of providing stable streams.
- **Cost** : Utilization of existing hardware, drastically reducing the initial investment.
- **Wireless Capability** :

IV. 3.2 EVALUATION OF STREAMING PROTOCOLS

To route the video streams from the smartphones to the processing computer, three main protocols were compared:

Protocol	Infra-structure	Latency	Stability / Cost
SRT	4G / 5G	Low	Excellent; high subscription costs
RTMP	4G/5G + remote server	Medium	Industry standard (Youtube, Twitch); requires a server
HTTP (Local)	Wi-Fi (router)	Optimal	Excellent stability; no recurring cost

The **HTTP protocol over a local network** was selected. Unlike cellular solutions (SRT/RTMP) that depend on public network load and incur data costs, the local network allows full control, high stability, and minimizes latency.

V. 3.3 LOCAL NETWORK IMPLEMENTATION AND TOPOLOGY

The network architecture put in place relies on the creation of a **high-performance private Wi-Fi network** (using a HUAWEI WiFi AX3-type router). This configuration makes it possible to avoid potential interference with the public networks of stadiums.

The system is structured as follows:

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- 1) **Wireless transmission:** The four smartphones transmit their respective streams to the central access point via the HTTP protocol. Tests in real conditions validated an effective range of **80 metres**, covering a significant portion of the playing surface.
- 2) **Downstream link:** The router is connected to the processing computer (YOLOv8 computing unit) via a gigabit Ethernet connection. This wired link ensures that the bottleneck does not occur at the final data reception stage.
- 3) **Centralised processing:** The computer simultaneously receives the four streams to perform player detection and tracking, enabling a precise spatial reconstruction of the game action.



Fig. 4. Network topology: four smartphones transmit wirelessly via HTTP (80m range) to a private Wi-Fi router, which forwards data via gigabit Ethernet to the YOLOv8 processing computer.

A. Camera score algorithm

Please use “soft” (e.g., $\text{eqref}\{Eq\}$) cross references instead

B. IHM

- The word “data” is plural, not singular.

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TABLE I
TABLE TYPE STYLES

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	Table column subhead	Subhead	Subhead
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^aSample of a Table footnote.

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Fig. 5. Example of a figure caption.

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ACKNOWLEDGMENT

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REFERENCES

- [1] G. Eason, B. Noble, and I. N. Sneddon, “On certain integrals of Lipschitz-Hankel type involving products of Bessel functions,” *Phil. Trans. Roy. Soc. London*, vol. A247, pp. 529–551, April 1955.
- [2] J. Clerk Maxwell, *A Treatise on Electricity and Magnetism*, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68–73.
- [3] I. S. Jacobs and C. P. Bean, “Fine particles, thin films and exchange anisotropy,” in *Magnetism*, vol. III, G. T. Rado and H. Suhl, Eds. New York: Academic, 1963, pp. 271–350.
- [4] K. Elissa, “Title of paper if known,” unpublished.
- [5] R. Nicole, “Title of paper with only first word capitalized,” *J. Name Stand. Abbrev.*, in press.
- [6] Y. Yorozu, M. Hirano, K. Oka, and Y. Tagawa, “Electron spectroscopy studies on magneto-optical media and plastic substrate interface,” *IEEE Transl. J. Magn. Japan*, vol. 2, pp. 740–741, August 1987 [Digests 9th Annual Conf. Magnetism Japan, p. 301, 1982].
- [7] M. Young, *The Technical Writer’s Handbook*. Mill Valley, CA: University Science, 1989.
- [8] Roboflow Universe public project made by Rugby Ruck. “Rugby Detection Computer Vision Model”. <https://universe.roboflow.com/rugby-ruck/rugby-detection-m0wvvs> . Accessed: 17 April 2026.

- [9] Roboflow. “Football AI Tutorial: From Basics to Advanced Stats with Python”. 22 August 2024. Accessed: April 2026. Available: <https://www.youtube.com/watch?v=aBVGKoNZQUw>

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